

Title

Activated vitamin D supplementation did not prevent acute kidney injury in a randomized trial of critically ill patients

Abstract

Background: While vitamin D metabolites modulate immune and renal function in vitro and in animal models, clinical translation remains uncertain. **Methods:** One hundred fifty ICU patients at baseline risk for acute kidney injury were randomly assigned to receive calcifediol (n=51), calcitriol (n=50), or placebo (n=49). The primary endpoint was a hierarchical composite (7-day mortality, renal replacement therapy requirement, creatinine elevation) scored using a rank-based algorithm. Secondary endpoints included acute kidney injury incidence and combined dialysis/death. **Results:** Primary endpoint rank scores were similar across groups: calcifediol versus placebo $p=0.85$; calcitriol versus placebo $p=0.58$. Secondary outcomes showed no between-group differences. One hypercalcemia case occurred in each active treatment group; none in placebo. Transcriptomic analysis revealed calcitriol-induced upregulation of immune and metabolic pathways (IFN signalling, oxidative phosphorylation, DNA repair) compared with calcifediol or placebo. **Conclusions:** Neither activated vitamin D formulation reduced acute kidney injury risk or mortality in this high-risk population, despite favorable immunological signalling patterns.